6 Constraint workflow

To conclude, there are three immediate directions. First, I will implement a practical Hamilton Path example inside an actual stage or Avoidance sequence. Second, I will improve the 3SAT example, especially its readability, pacing, difficulty, and synchronization with music. Third, I will evaluate both logical correctness and playability. The central question is whether formal constraints can help us create expressive bullet hells without losing a guaranteed route. These prototypes are only the starting point, and I would appreciate feedback on both the formal model and the game design.

7 3SAT prototype

This is the current 3SAT prototype. On the left, the player assigns a truth value to each variable by moving into the TRUE or FALSE region before the timer ends. On the right, the game verifies a clause. Each green label represents a literal that is satisfied by the current assignment. A clause is passable when at least one of its three literals is true. In this way, a Boolean assignment becomes a sequence of movement decisions, and clause satisfaction becomes a playable obstacle check.

8 Why the mapping matters

The key research claim is correspondence, not only visual similarity. A variable becomes a forced TRUE or FALSE choice. A literal becomes a condition linked to that choice. A clause becomes a pass-or-fail gate, and a satisfying assignment becomes a complete sequence that can survive every gate. If this mapping is correct, the stage is clearable exactly when the represented formula is satisfiable. Establishing and testing this relationship will be an important part of the research.

9 Live 3SAT demonstration

Switch to the game. Show one timed variable assignment, then show how the clause checks respond. Intentionally demonstrate a failing assignment once; replay with a satisfying assignment and clear the sequence. Ask the audience to notice whether the relationship between the logical choice and the game response is understandable. Then return to the slides.

10 Hamilton Path prototype

The second prototype uses Hamilton Path. Each yellow platform represents a vertex, and permitted transitions represent graph edges. The player must visit every platform exactly once and then reach the green goal. Revisiting a platform or taking a transition that is not represented by an edge should be prevented or punished. A valid Hamilton path therefore supplies a complete safe route through the stage. At present, this is a provisional prototype; integrating it into an actual barrage is one of the next tasks.

11 Closing and next steps

To conclude, there are three immediate directions. First, I will implement a practical Hamilton Path example inside an actual stage or Avoidance sequence. Second, I will improve the 3SAT example, especially its readability, pacing, difficulty, and synchronization with music. Third, I will evaluate both logical correctness and playability. The central question is whether formal constraints can help us create expressive bullet hells without losing a guaranteed route. These prototypes are only the starting point, and I would appreciate feedback on both the formal model and the game design.

Speaking cue: pause after each key claim; emphasize the words in the slide title.